

20598 – Finance with Big Data

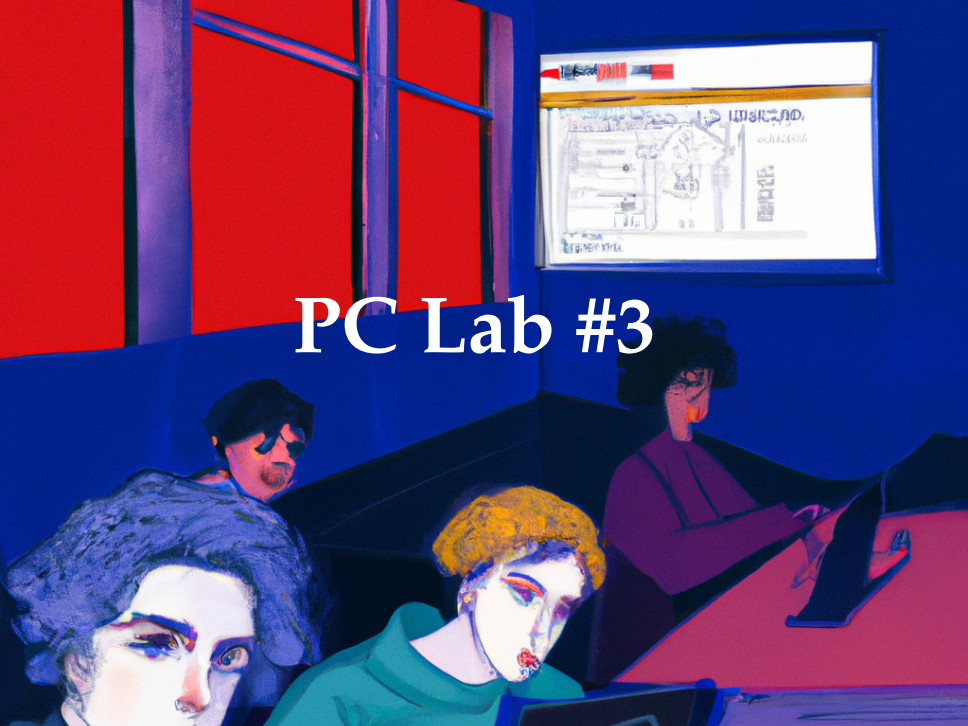
PC Lab #3: Creating a Factor from Text Data (Week 4)

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PC Lab #3



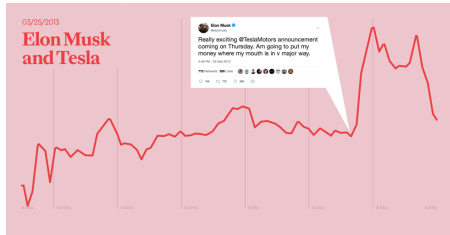
PC Labs Grading

- PC Labs solutions are submitted as Jupyter Notebooks, via email
 - Email title : PCLab#3 - Group X - Name1 Name2 Name3
 - Your Jupyter Notebook starts in the same way (same .ipynb name)
 - Tell me (again) how long did it take
- PC Labs grade will depend on :
 - Your ability to submit it **before the deadline (Friday, midnight)**
 - The **quality** of your code (comments, readability, use of functions, etc.)
 - The **structure** of the Jupyter Notebook: well organized, explain what you are doing and why
 - Your ability to **complete the tasks** and **innovate**
 - You should maybe produce less, but more *useful* outputs

Goals

- Manipulate and visualize financial social media messages
- Clean text data
- Perform Tweets' sentiment analysis
- Compute a measure of media attention

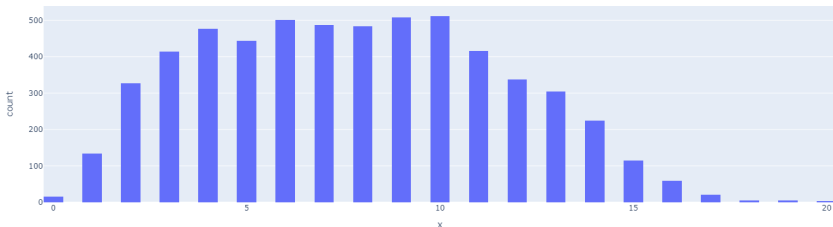
Prices are plummeting
because everyone's selling!
What should we do?



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Task #1: Basic manipulation and descriptive statistics

- Import the `StockTweetData1-2.csv` data and describe the sample (data available on BBoard)
- Data from [Stocktwits](#), the world's largest social network for investors and traders — be careful, data isn't clean !
- How many tweets, how many words per tweets, distribution of number of words per tweets, average sentiment, etc.



Task #2 : Cleaning and Visualization

(You might know better than me what to do here)

- Usual **cleaning steps**:
 - cleaning URLs, mentions, hashtags, emojis — no hashtags but \$ signs
 - tokenization, lemmatization, stopwords removal
 - tweet-specific text preprocessing (normalizes hashtags, elongated words, emoticons).
- What is the number of unique words?
- Plot a **word cloud** for text with positive and negative sentiment separately (see later for sentiment)

Task #3 : Sentiment analysis

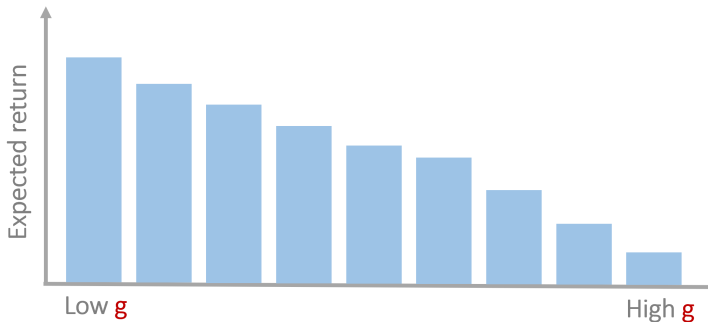
- To produce your own sentiment analysis, use either :
 - Pretrained transformer (e.g., Twitter-RoBERTa, FinBERT)
 - GPT-style LLM zero-shot/few-shot classification
- What is the performance of these methods on financial tweets ? Which one would you choose and why ?

Task #4 : Measuring media attention (1)

- Use the list of tickers gathered during last PC Lab (see the web-scraping part) to compute the number of tweets about each stock for year t :
 - e.g., AAPL: X tweets, Y negative, Z positive, N neutral
- Rank the stocks by their amount of total media attention, or, alternatively : positive and negative media attention, level of disagreement (dispersion), etc.
- Be creative !

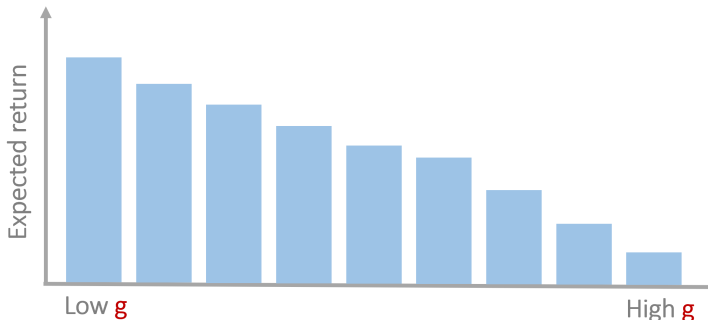
Task #4 : Measuring media attention (2)

- Create 10 portfolios based on your preferred measure of media attention
- Do you see a correlation between media attention g_t and stock returns at t ? It can be yearly, monthly, daily.



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- If yes, could Twitter attention is likely to be a good factor?
- Optional : same, but use [FF-5](#) factors to purge returns from what a traditional model predicts, and plot media attention vs. errors

Task #5 : Measuring media attention (3) — Optional

- Download the Fama–French 5 factors (Mkt-RF, SMB, HML, RMW, CMA, and RF) from [Ken French's Data Library](#)
- Regress the stock's excess returns on these five factors
- Save the residuals from this regression. These represent the component of returns unexplained by traditional risk factors (abnormal returns).
- Redo same plot (see previous slide), but residuals vs. media attention.